



MINDS. JWST-MIRI Observations of a Spatially Resolved Atomic Jet and Polychromatic Molecular Wind toward SY Cha

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Abstract

The removal of angular momentum from protostellar systems drives accretion onto the central star and may drive the dispersal of the protoplanetary disk. Winds and jets can contribute to removing angular momentum from the disk, though the dominant process remains unclear. To date, observational studies of resolved disk winds have mostly targeted highly inclined disks. We report the detection of extended H₂ and [Ne II] emission toward the young stellar object SY Cha with the JWST Mid-Infrared Instrument Medium-Resolution Spectrometer (MIRI-MRS). This is one of the first polychromatic detections of extended H₂ toward a moderately inclined, $i = 51^\circ$, Class II source. We measure the semiopening angle of the H₂ emission and build a rotation diagram to determine the H₂ excitation temperature and abundance. We find a wide semiopening angle, high temperature, and low column density for the H₂ emission, all of which are characteristic of a disk wind. We derive a molecular wind mass loss rate of $3 \pm 2 \times 10^{-9} M_\odot \text{ yr}^{-1}$, which is high compared to the previously derived stellar accretion rate of $6.6 \times 10^{-10} M_\odot \text{ yr}^{-1}$. This suggests either that the stellar accretion and the disk wind are driven by different mechanisms or that accretion onto the star is highly variable. These observations demonstrate MIRI-MRS's utility in expanding studies of resolved disk winds beyond edge-on sources.

Unified Astronomy Thesaurus concepts: [Protoplanetary disks \(1300\)](#); [Infrared spectroscopy \(2285\)](#); [Molecular gas \(1073\)](#); [Stellar jets \(1607\)](#)

1. Introduction

Stars form from collapsing clouds of gas and dust. Due to conservation of the angular momentum, not all of the material is able to collapse directly onto the star, instead forming a protoplanetary disk. In order for material to accrete from the disk onto the star it must first shed its angular momentum. Turbulence-driven viscous spreading can redistribute angular momentum within the disk (N. I. Shakura & R. A. Sunyaev 1973). Alternatively, outflows can remove angular momentum from the system entirely (R. E. Pudritz & C. A. Norman 1983). Outflows also drive the dissipation of the protoplanetary disk.

Outflows are variously classified as winds and jets based on the degree of collimation and outflow velocity. Winds launched by the disk can be driven either by high-energy radiation from the central star—photoevaporative winds—or by the stellar and/or disk magnetic field—magnetohydrodynamic (MHD) winds; see I. Pascucci et al. (2023) for a recent review. Photoevaporative winds are expected to be launched from radii greater than a tenth of the gravitational radius but small enough for the disk to intercept a large amount of high-energy radiation from the central star, and to be relatively slow, with velocities $\leq 10 \text{ km s}^{-1}$ (D. Hollenbach et al. 1994; K. Liffman 2003; C. J. Clarke & R. D. Alexander 2016). In contrast, MHD winds can be launched from a large range of radii and have larger initial velocities and thus larger vertical extents (R. D. Blandford & D. G. Payne 1982; G. R. J. Lesur 2021). Which of these processes is the dominant form of mass removal remains unclear (L. Hartmann et al. 2016). Jets are both more



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Table 1
Properties of SY Cha

Parameter	Value	References
Stellar mass	$0.70 M_{\odot}$	(1, 2, 3)
Stellar luminosity	$0.55 L_{\odot}$	(1, 2, 3)
Effective temperature	4030 K	(1, 2, 3)
Accretion luminosity	$7.39 \times 10^{-3} L_{\odot} \text{ yr}^{-1}$	(1, 2, 3)
Mass accretion rate	$6.6 \times 10^{-10} M_{\odot} \text{ yr}^{-1}$	(1, 2, 3)
Spectral type	K5Ve	(4)
A_v	0.5	(5)
Distance	180.7 pc	(3)
Disk inclination	51.1°	(6)

References: (1) C. F. Manara et al. (2016); (2) G. A. Feiden (2016); (3) Gaia Collaboration et al. (2021); (4) A. Frasca et al. (2015); (5) C. F. Manara et al. (2016); (6) R. Orihara et al. (2023).

collimated (semiopening angle $< 20^{\circ}$) and have a higher velocity ($\geq 100 \text{ km s}^{-1}$) than winds (A. Frank et al. 2014). They are launched via MHD processes in the innermost regions of the disk or at the stellar surface, though the exact details are still uncertain (Y. Tsukamoto et al. 2023).

Forbidden line emission from atomic ions can trace both winds and jets (I. Pascucci et al. 2007, 2020; R. van Boekel et al. 2009; I. Pascucci et al. 2020; L. Flores-Rivera et al. 2023; N. S. Bajaj et al. 2024; S. A. Federman et al. 2024; Ł. Tychoniec et al. 2024). Extended emission from rovibrational as well as pure rotation lines of H_2 has been observed toward a number of young stellar objects and is often attributed to either an MHD disk wind or shocks in the outflow cavity (T. L. Beck et al. 2008; S. Melnikov et al. 2023; N. Arulanantham et al. 2024; V. Delabrosse et al. 2024; Ł. Tychoniec et al. 2024).

Here we present multiline analysis of extended H_2 emission toward a Class II protoplanetary disk. SY Cha hosts a moderately inclined transition disk, displaying a 70 au cavity seen both in millimeter continuum emission (R. Orihara et al. 2023) and near-infrared scattered light (B. B. Ren et al. 2023; C. Ginski et al. 2024; S. Juillard et al. 2024). A summary of the stellar and disk properties is given in Table 1. The JWST observations were taken as part of the MIRI Mid-Infrared Disk Survey (MINDS). For a description of the full sample and observing strategy, see I. Kamp et al. (2023) and T. Henning et al. (2024).

2. Observations and Data Reduction

Observations of SY Cha with the Mid-InfraRed Instrument (MIRI; G. H. Rieke et al. 2015; G. S. Wright et al. 2015) Medium-Resolution Spectrometer (MRS; M. Wells et al. 2015) were carried out on 2022 August 8 as part of the MINDS GTO Program (PID: 1282, PI: T. Henning). As the source is bright, dedicated target acquisition was not carried out. A four-point dither pattern in the positive direction was used, and the total exposure time was 3696 s. Data reduction was carried out following the process described by K. R. Schwarz et al. (2024), though with an updated version (1.13.3) of the JWST pipeline (H. Bushouse et al. 2023). Visual inspection of the resulting data cubes reveals extended emission at wavelengths corresponding to the H_2 and [Ne II] lines. Here we focus on this extended emission. For information on the spatially unresolved continuum and line emission, see K. R. Schwarz et al. (2024).

The extended H_2 emission is not easily seen in the spectral images directly because it is comparable in surface brightness to the wings of the point-spread function (PSF). However, after the subtraction of the adjacent continuum spectral images the extended emission is revealed (see Appendix A), indicating an outflow structure. In order to sharpen these H_2 images we apply a deconvolution method, the Lucy–Richardson algorithm (W. H. Richardson 1972; L. B. Lucy 1974). This requires the PSF for each H_2 line, but instead of using theoretical templates or observations of calibration point sources we simply use the continuum spectral images. The advantage is the identical dithering and preprocessing, although we cannot be sure that the source is point like in the continuum. The number of adjacent spectral images is a trade-off between the signal-to-noise ratio (S/N) and how closely the deduced “PSF” will represent that of the H_2 line wavelength. As a compromise we chose the median of 30 spectral images at each side of the emission line, which is represented by the coadded 5 spectral images covering the line. By increasing the number of iterations, the features get sharper (Appendix A), and how far one can proceed depends basically on the S/N. The observed H_2 lines differ in this respect, and as a compromise we chose nine iterations (Figure 1). These images are used in all following analysis unless otherwise noted. The $S(3)$ – $S(7)$ lines all appear spatially extended. The [Ne II] emission also appears spatially extended (Figure 2). The excess blue- and redshifted [Ne II] emission after PSF and continuum subtraction is shown in Figure 2. Deconvolution is not performed on the [Ne II] emission. [Ne III] and [Ar II] are detected but do not appear extended. No other forbidden lines are detected. Notably, there is no evidence of the [Fe II] line at $17.936 \mu\text{m}$, though a weak feature could be masked by overlapping water emission (K. R. Schwarz et al. 2024).

3. Analysis

The [Ne II] emission originates primarily from the region around the central source. Additionally, extended emission is clearly seen on either side of the central source, roughly perpendicular to the protoplanetary disk (R. Orihara et al. 2023; C. Ginski et al. 2024). This extended emission is strongest in the spectral images adjacent to the central [Ne II] frame at $12.814 \mu\text{m}$ (see Appendix B). The blue- and redshifted emission is binned with a velocity offset of 117 km s^{-1} , extending roughly $3''$ from the central source in opposite directions. The blueshifted emission appears brighter, as expected for a bipolar outflow where the light from the redshifted lobe is subject to more extinction along the line of sight. Following the method of I. Pascucci et al. (2025), we determine the semiopening angle of the blue- and redshifted [Ne II] emission by fitting a Gaussian profile to cuts perpendicular to the outflow axis. This allows us to derive a FWHM. At this wavelength the pixel scale is $0.2''$. Any derived FWHM below this value is considered unresolved and not used to determine the semiopening angle. We then derive the semiopening angle from the slope of line fit to the remaining data points. The derived semiopening angle for the blueshifted lobe is $6.3 \pm 1.8^{\circ}$, consistent with measurements for jets in other sources (N. Arulanantham et al. 2024; V. Delabrosse et al. 2024; I. Pascucci et al. 2025).

The H_2 $S(3)$ – $S(7)$ extended emission primarily originates from the nearside of the disk, the same side as the blueshifted [Ne II]. There are also hints of extended emission on the farside of disk, most clearly seen in the $S(5)$ line. The H_2 emission appears less collimated than the [Ne II] emission. The higher-

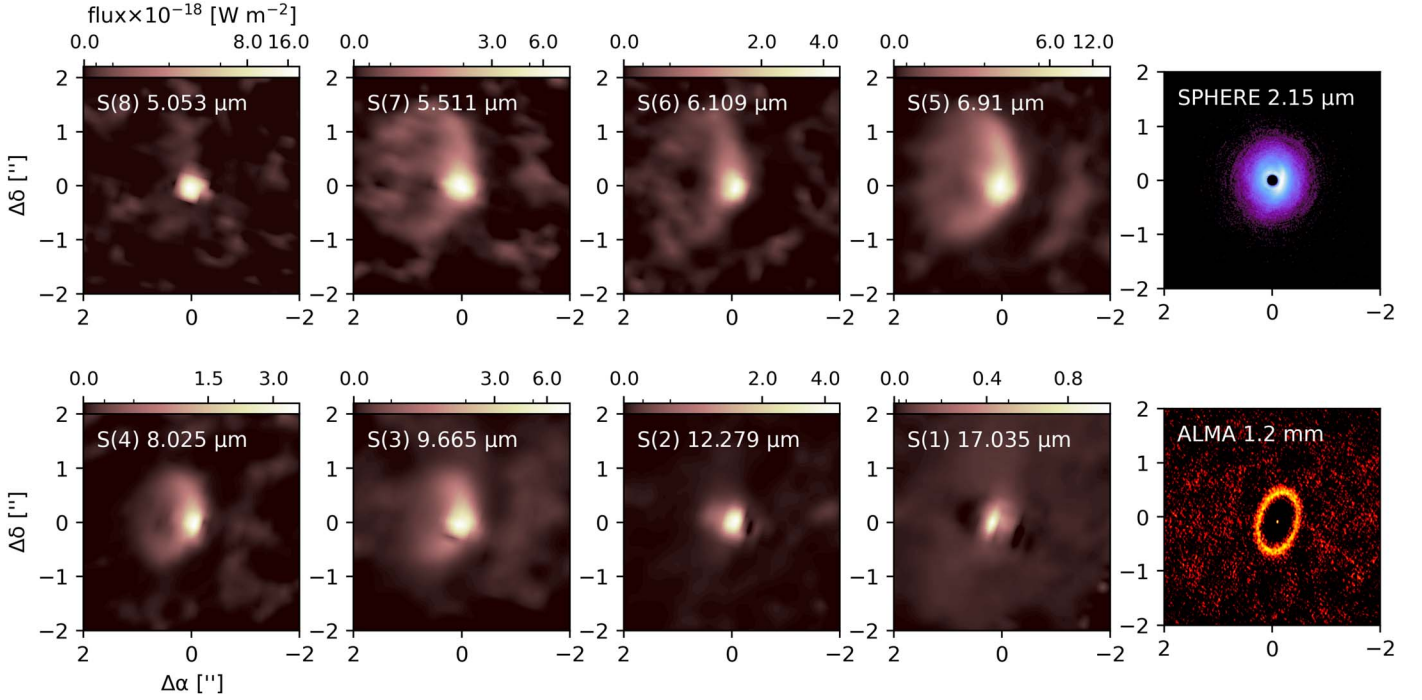


Figure 1. H_2 lines after nine rounds of deconvolution. The right panels show the scattered light emission observed by SPHERE and imaged by S. Juillard et al. (2024) and the dust continuum emission observed with the Atacama Large Millimeter/submillimeter Array (R. Orihara et al. 2023).

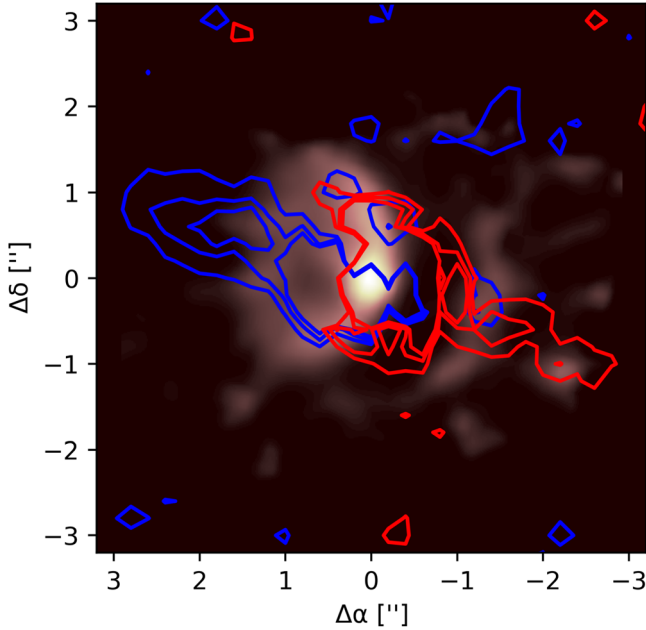


Figure 2. Blue- and redshifted wings of the [Ne II] emission, with an offset of 117 km s^{-1} . Background is the $S(5)$ H_2 line.

energy transitions generally appear more extended than the lower-energy transitions: the 3σ contour for $S(7)$ in Figure 1 extends $1''.78$ perpendicular to the plane of the protoplanetary disk and $1''.35$ horizontally, while for $S(3)$ it is only $0''.99$ and $0''.96$. This difference could be due to the lower signal to noise at longer wavelengths.

To determine the semiopening angle of the H_2 emission we first identify the offset of the brightest pixel at each position along the x -axis (Figure 3). This vertical offset is deprojected assuming the inclination of the H_2 emission is the same as the disk inclination. We then overlay the two sides before fitting a

straight line and deriving the semiopening angle from our fit, as seen in the bottom panel of Figure 3. The derived semiopening angles range from $64^\circ \pm 2^\circ$ for $S(7)$ to $48^\circ \pm 1^\circ$ for $S(5)$. It should be noted that pixel scale of the MRS increases from $0''.196$ in Channel 1 to $0''.273$ in Channel 4. Additionally, the FWHM of the PSF is known to increase with wavelength (I. Argüriou et al. 2023; C. J. Law et al. 2023).

3.1. Molecular Hydrogen Rotation Diagram

The H_2 emission could potentially originate from a wind or from the hot surface of the disk. For a population of molecules in LTE, the number of molecules at each upper state excitation level, N_u , is given by

$$N_u = \frac{N}{Z} g_u \exp(-E_u/kT), \quad (1)$$

where N is the total column density, $Z(T)$ is the partition function for a given excitation temperature T , g_u and E_u are the statistical weight and upper state energy, and k is the Boltzmann constant. Assuming optically thin emission, the observed flux F_u is related to the upper state column density:

$$N_u = \frac{4\pi F_u}{A_u h\nu\Omega}, \quad (2)$$

where Ω is the emitting area and A_u is the Einstein A rate coefficient for spontaneous emission. Equation (1) can then be rewritten as

$$\ln \frac{N_u}{g_u} = \ln \frac{4\pi F_u}{g_u A_u h\nu\Omega} = -\frac{E_u}{kT} - \ln \frac{Z(T)}{N}. \quad (3)$$

We extract the flux for each line by summing the flux within $0''.5$ and $2''$ radius apertures centered on the source, allowing us to differentiate between emission at the central source and extended emission. The values used in constructing the rotation

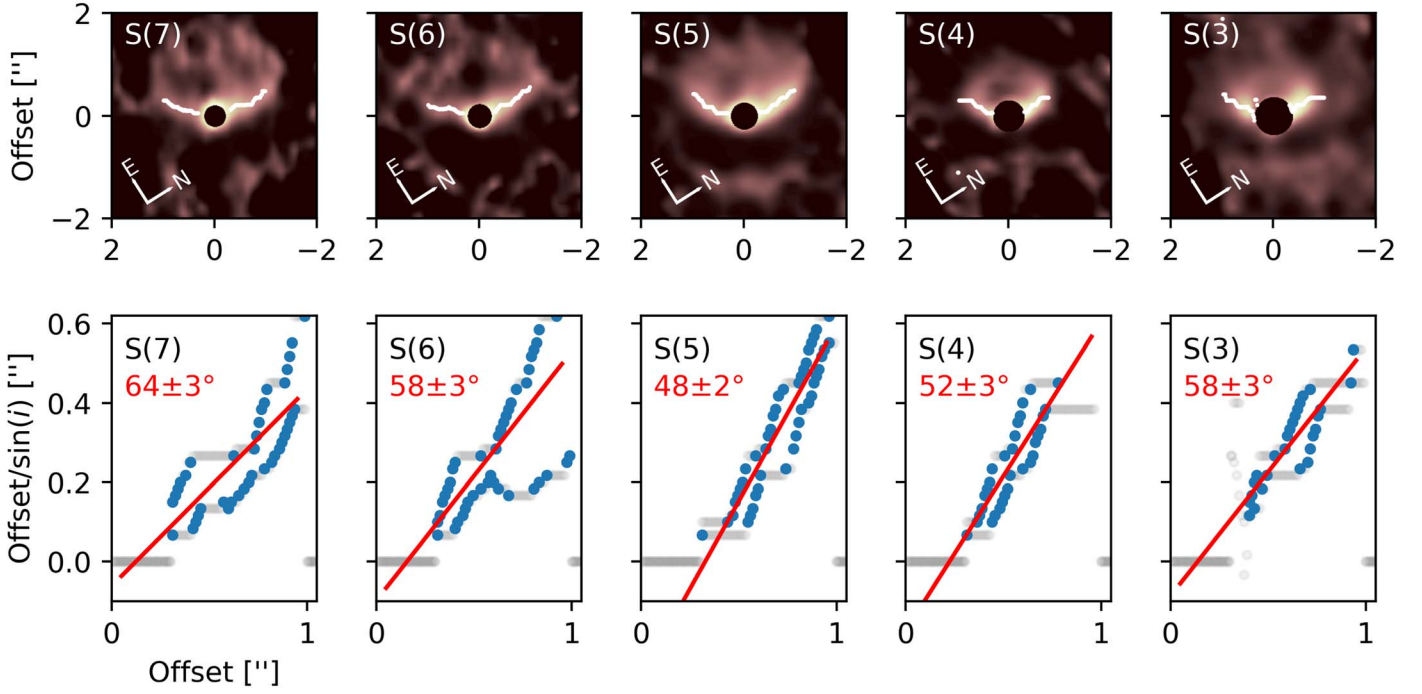


Figure 3. Top: white points indicate the brightest pixel at each position along the x -axis for the five H_2 lines with extended emission. The central region inside the diffraction limit is masked to better show the faint extended emission. Bottom: best-fit line (red) tracing the extended emission. For a given vertical offset only the point with the smallest horizontal offset is used in the fit (blue points). Points not used in the fit are shown in gray. The derived semiopening angle for each line is listed in red.

Table 2
 H_2 Line Data

Transition	Line Center (μm)	E_u (K)	g_u	n_{crit} (cm^{-3})	Flux Density $\times 10^{-4}$ ($\text{erg s}^{-1} \text{cm}^{-2} \text{str}^{-1}$)		
					($0''\text{--}2''$)	($0''\text{--}0.5''$)	($0.5''\text{--}2.0''$)
S(1)	17.0348	1015.1	21	$2.4\text{e}2$	0.74 ± 0.04	3.2 ± 0.2	0.58 ± 0.05
S(2)	12.2786	1681.6	9	$1.7\text{e}3$	1.10 ± 0.06	12.7 ± 0.7	0.33 ± 0.08
S(3)	9.66491	2503.7	33	$5.5\text{e}3$	3.6 ± 0.2	27 ± 1	2.0 ± 0.2
S(4)	8.02504	3474.5	13	$2.0\text{e}4$	1.38 ± 0.08	11.3 ± 0.6	0.7 ± 0.1
S(5)	6.90951	4586.1	45	$3.9\text{e}4$	7.4 ± 0.4	60 ± 3	3.9 ± 0.5
S(6)	6.10856	5829.8	17	$1.0\text{e}5$	2.2 ± 0.1	16.6 ± 0.9	1.2 ± 0.1
S(7)	5.51118	7196.7	57	$1.5\text{e}5$	4.4 ± 0.3	33 ± 2	2.5 ± 0.3
S(8)	5.05306	8677.1	21	...	2.6 ± 0.1	30 ± 2	0.7 ± 0.2

Note. E_u and g_u values from E. Roueff et al. (2019). n_{crit} for collisions with H and a temperature of 1000 K from J. Le Bourlot et al. (1999).

diagram are given in Table 2, where the uncertainty is propagated from the rms noise and added in quadrature with the 5% absolute flux calibration uncertainty (I. Argyriou et al. 2023). We do not correct for extinction along the line of sight as the visual extinction toward SY Cha is low (A_V of 0.5; C. F. Manara et al. 2016) and the uncertainty in converting to an IR optical depth is large (M. McClure 2009).

We determine the column density N by using our best-fit excitation temperature T to solve for the partition function Z . Here we use the normal H_2 partition function of A. Popovas & U. G. Jørgensen (2016), which treats the ortho- and para- H_2 separately. We convert the derived column density into a total H_2 mass assuming that the emitting area is equal to our aperture; $2''$ is an overestimate of the true emitting area and thus results in an upper limit on the true H_2 mass. The resulting values are given in Table 3.

Table 3
 H_2 Properties Derived from Rotation Diagram

Parameter	$2.0''$	$0.5''$	$0.5''\text{--}2.0''$
T_{hot} [K]	1700 ± 300	1300 ± 500	1260 ± 90
T_{warm} [K]	600 ± 100	600 ± 300	...
N_{hot} [cm^{-2}]	$1.5 \pm 1.0 \times 10^{18}$	$1 \pm 2 \times 10^{19}$	$1.1 \pm 0.3 \times 10^{18}$
N_{warm} [cm^{-2}]	$2.0 \pm 1.0 \times 10^{19}$	$1 \pm 1 \times 10^{20}$	...
$M_{\text{hot}}(H_2)$ [$M_{\odot}$]	$4 \pm 3 \times 10^{-7}$	$2 \pm 4 \times 10^{-7}$	$3.1 \pm 0.9 \times 10^{-7}$
$M_{\text{warm}}(H_2)$ [$M_{\odot}$]	$7 \pm 4 \times 10^{-6}$	$2 \pm 2 \times 10^{-6}$	...

There is a clear change of slope in the rotation diagram in Figure 4, necessitating a two-component fit. The hot component is fit by a temperature of 1700 ± 300 K and a column density of $6.0 \pm 0.1 \times 10^{19} \text{cm}^{-2}$. The warm component is fit by a temperature of 600 ± 100 K, consistent with emission originating

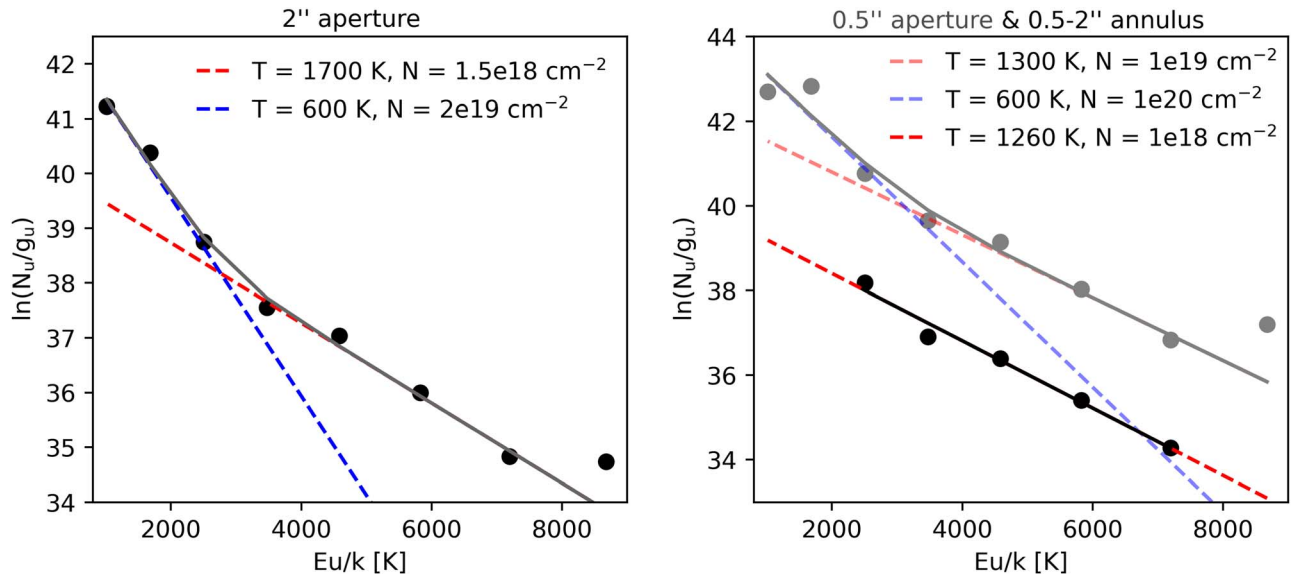


Figure 4. Rotation diagrams with a two-component fit. The left panel shows the fit for all fluxes within a 2'' radius aperture centered on the source. The right panel shows the fit for the flux within a 0.5'' aperture (gray) and the flux within an annulus between 0.5'' and 2.0'' (black).

from the disk surface (e.g., R. Franceschi et al. 2024), and a column density of $2 \pm 1 \times 10^{19} \text{ cm}^{-2}$ (Table 3). It is also clear from Figure 4 that the lines for which we detect extended emission ($E_u = 2503.7\text{--}7196.7 \text{ K}$) have a large contribution from the hot component. To test if the two temperature components are due to two distinct origins, rather than a gradual temperature gradient, we redo our analysis on only the emission within the inner 0.5'' and on only the emission from an annulus from 0.5'' to 2'' (Figure 4, right panel). For the inner region, we exclude the 8677.1 K point from the fit as this results in an unrealistically high temperature for the hot component. As for the full 2'' region, fitting the inner 0.5'' requires a two-component fit. For the outer annulus, we only consider those lines for which extended emission is observed. For this region a single-component fit is sufficient. T_{warm} for the inner 0.5'' is the same as for the full region. T_{hot} in the inner and outer regions agrees within the uncertainty, 1300 K, though both values are several hundred Kelvin lower compared to the full 2''. The higher temperature for the full aperture measurement is due to the inclusion of the $S(8)$ line. Excluding this line results in $T_{\text{warm}} = 1300 \pm 200 \text{ K}$. The overall consistency between the derived temperatures suggests two distinct origins: one limited to the inner regions and characterized by a temperature of 600 K, likely tracing the hot disk surface, and one characterized by a temperature of 1300–1700 K, likely an outflow.

We can use the derived column density for the hot component to check if our assumption of LTE is valid. To convert from an H_2 column density to an average number density we divide the column density by the observed extent of the emission on the sky of 1.78. This results in an average H_2 number density of $n(\text{H}_2) = 228 \text{ cm}^{-3}$. In this high temperature, a low-density regime is expected to be in the form of neutral H or H^+ , with $n(\text{H})$ more than a factor of 100 greater than $n(\text{H}_2)$ (E. Bron et al. 2016; R. Nakatani et al. 2018; A. D. Sellek et al. 2024). This makes atomic hydrogen the main collisional partner with a density of $n(\text{H}) \sim 2 \times 10^4 \text{ cm}^{-3}$. According to the critical densities for collisions with H atoms at 1000 K given in Table 2, only the lowest-energy lines for which we observe extended emission are expected to be in LTE (J. Le

Bourlot et al. 1999). LTE assumes the population distribution for energy levels of a given species can be described by a single excitation temperature. Using LTE assumptions for a population in non-LTE can result in an underestimate of the derived column density by more than 1 order of magnitude (J. G. Mangum & Y. L. Shirley 2015). As such, the column density reported here should be considered a lower limit. More advanced modeling-based analysis is needed to obtain more precise measurements (e.g., F. F. S. van der Tak et al. 2007; C. Brinch & M. R. Hogerheijde 2010).

3.2. Atomic Ions

We determine the total flux from [Ne II], [Ne III], and [Ar II] within the central aperture, which contains primarily emission from the disk, by fitting a Gaussian to each line. As the [Ne III] overlaps with a nearby water line, two Gaussians are fit, though only the Gaussian centered on the [Ne III] is used to determine the flux; see Appendix C. Analysis of ions within the disk can be used to constrain the high-energy component of the stellar radiation field impinging on the disk, which in turn can drive the disk wind. In particular, the ratio of [Ne II]/[Ne III] differentiates between ionization from hard EUV photons and soft EUV and/or X-rays, while [Ne II]/[Ar II] differentiates between hard X-rays and soft X-rays or EUV (D. Hollenbach et al. 2009). However, for SY Cha the [Ne II] emission clearly includes a substantial jet component, which can be driven by locally produced UV emission from shocks (e.g., D. Hollenbach 1997) and is thus not a reliable tracer of the stellar radiation field. Higher-spectral-resolution observations (e.g., I. Pascucci et al. 2020) would be needed to separate out the jet emission from any disk emission.

4. Discussion

4.1. Extended Molecular Hydrogen Emission

C. Ginski et al. (2024) recently analyzed scattered light observations of SY Cha at 1.6 and 2.2 μm . They derive a disk surface height at 72 au of 7–12 au. This is significantly closer to the disk midplane than the rotational H_2 emission, which has a

height of 35 au at a radius of 72 au assuming an opening angle of 64° . This suggests the H_2 does not originate from the disk surface. This can also be seen in Figure 1, where the scattered light surface as imaged by SPHERE is noticeably flatter than the H_2 surface. Additionally, our rotation diagram analysis finds an excitation temperature of 1260 ± 90 K for the lines seen in extended emission, much hotter than the expected disk warm molecular layer at large radii (C. J. Law et al. 2023; T. Paneque-Carreño et al. 2023). Therefore the extended H_2 emission is likely tracing an outflow.

Previous studies of JWST-MIRI extended H_2 emission in similar systems report narrower semiopening angles of 9° – 35° and similar to higher temperatures, attributed to a scattered MHD disk wind (N. Arulanantham et al. 2024; V. Delabrosse et al. 2024). These systems are more inclined than SY Cha, and the extended H_2 was observed in the $S(2)$ and $S(1)$ transitions, which do not appear extended for SY Cha. Interestingly, V. Delabrosse et al. (2024) find the redshifted wind in DG Tau B has a very narrow semiopening angle, while the semiopening angle of the blueshifted emission is much wider. I. Pascucci et al. (2025) used JWST-NIRSpec to resolve H_2 disk winds in the $S(9)$ transition for four edge-on disks. They find semiopening angles ranging from 13° to 50° for the blueshifted wind. The semiopening angle for Tau 042021 in $S(9)$ is 38.5 ± 4.5 and 37.66 ± 2.6 for the two lobes of the blueshifted outflow, consistent with the value of 35 ± 5 for $S(2)$ found by N. Arulanantham et al. (2024) using MIRI observations. Based on our derived temperature and semiopening angle, we conclude the extended H_2 emission in SY Cha is likely from a disk wind. This analysis of a spatially extended molecular wind toward a moderately inclined protoplanetary disk, along with several previous studies (e.g., M. Güdel et al. 2018; A. S. Booth et al. 2021; V. Delabrosse et al. 2024), demonstrates that studies of resolved disk winds need not focus on edge-on systems.

The 600 K temperature of the warm H_2 component is consistent with a disk origin. However, the derived warm H_2 column density of $6.0 \pm 0.1 \times 10^{19} \text{ cm}^{-2}$ is significantly smaller than the H_2 column density of $5 \times 10^{25} \text{ cm}^{-2}$ for the very low-mass star 2MASS-J16053215-1933159 derived by R. Franceschi et al. (2024). This could be due to the shorter evolutionary times for disks around very low-mass stars, resulting in higher amounts of dust settling and more observable gas above the optically thick dust layer (P. Pinilla et al. 2013). The idea that disks around low-mass stars evolve more quickly has previously been used to explain the abundance of gas-phase hydrocarbons observed toward low-mass sources (B. Tabone et al. 2023; A. M. Arabhavi et al. 2024).

4.2. Wind Mass-loss Rate

Assuming a typical wind velocity of 10 km s^{-1} (R. Alexander et al. 2014; B. Tabone et al. 2020) we can convert our derived hot H_2 mass into a molecular wind mass-loss rate following the method of N. Arulanantham et al. (2024). We assume a maximum height above the midplane of 271 au based on the extent of the $S(5)$ emission in Figure 1 and assume an inclination angle of 51.1° . This results in a wind mass-loss rate of $3 \pm 2 \times 10^{-9} M_\odot \text{ yr}^{-1}$, in line with most reported molecular wind mass-loss rates for photoevaporative winds from low-mass Class II objects (F. Louvet et al. 2018; B. Tabone et al. 2020; A. R. Anderson et al. 2024; N. Arulanantham et al. 2024)

and lower than the mass-loss rate inferred for Class 0/I sources (C.-F. Lee et al. 2018; A. de Valon et al. 2020). The derived H_2 mass-loss rate is 1 order of magnitude greater than the mass accretion rate onto the central star reported in Table 1. The molecular wind component could be only a fraction of the total mass in the wind, with a significant fraction of the mass in neutral and ionized atomic gas (R. Nakatani et al. 2018; L. Wang et al. 2019; A. Komaki et al. 2021; A. D. Sellek et al. 2024). This would further increase the difference between the wind mass-loss rate and the stellar accretion rate. Our derived H_2 mass-loss rate puts SY Cha at the high end of the typical $\dot{M}_{\text{wind}}/\dot{M}_{\text{acc}}$ range (M. Fang et al. 2018; I. Pascucci et al. 2023). MHD winds are driven by accretion energy, while photoevaporative winds are thermally driven and do not transport significant angular momentum (G. Lesur et al. 2023). Thus, photoevaporative wind mass-loss rates are not physically linked to stellar accretion. The high ratio between our derived H_2 mass-loss rate and the stellar accretion rate favors the photoevaporative scenario. Another possibility is that SY Cha's stellar accretion rate is highly variable and was measured at a particularly low point. This would also explain the mid-IR variability between the three epochs of Spitzer observations and the JWST observations reported by K. R. Schwarz et al. (2024).

4.3. Extended [Ne II] Emission

The [Ne II] emission is spatially resolved, with both a blueshifted and redshifted component. The [Ne II] emission is both more extended and more collimated than the extended H_2 emission. Similar results from JWST-MIRI were recently reported for the edge-on source Tau 042021, where the jet was also seen in [Ne III], [Ni II], [Fe II], [Ar II], and [S III] (N. Arulanantham et al. 2024). These lines do not appear to be spatially extended in our observations of SY Cha, though this may be due in part to the weaker line emission. The undetected lines such as [Ni II], [Fe II], and [S III] could also be masked by stronger water lines in the central region.

The [Ne II] emission with a small velocity offset appears largely unresolved and cospatial with the central source, while there are also clearly two lobes of emission extending away from the disk corresponding to velocity offsets of greater than 100 km s^{-1} . High-spectral-resolution observations of [Ne II] toward other sources often show either a high-velocity component tracing a jet for sources with high accretion rates or a low-velocity component tracing a wind for low accretors (e.g., I. Pascucci et al. 2020). Recent JWST-MIRI-MRS observations of [Ne II] in edge-on disks are likewise either attributed to a disk wind (e.g., A. R. Anderson et al. 2024; N. S. Bajaj et al. 2024) or a jet (N. Arulanantham et al. 2024). For SY Cha, only the [Ne II] emission with a large velocity offset appears spatially extended. As the [Ne II] emission is extended and collimated, the wind origin seems to be excluded. This nested morphology, with H_2 tracing a wide angle wind and atomic forbidden line emission tracing a more collimated jet, has been seen in several sources (I. Pascucci et al. 2025). We conclude that while there may be a spatially unresolved wind component to the [Ne II] emission close to the central star, the dominant extended component comes from a jet. SY Cha is unusual in that it appears to be a low accretor with a [Ne II] jet.

5. Conclusions

We report spatially extended emission surrounding SY Cha observed with MIRI-MRS in the pure rotational $S(3)$ – $S(7)$ H_2 lines and [Ne II], as well as unresolved detections of [Ne III] and [Ar II]. Our conclusions are as follows:

1. The [Ne II] emission is highly collimated, with both blue- and redshifted lobes. This, as well as a narrow semiopening angle of $6.3 \pm 1.8^\circ$, is indicative of a jet.
2. The H_2 emission is seen primarily on the nearside of the source.
3. The semiopening angle of the H_2 is much wider than the [Ne II], ranging from $48^\circ \pm 1^\circ$ to $64^\circ \pm 3^\circ$ depending on the line.
4. The H_2 rotation diagram is best described by a two-component fit. The lines for which we see extended emission are all fit by a temperature of 1260 ± 90 K and a column density of $1.1 \pm 0.3 \times 10^{18} \text{ cm}^{-2}$, while the lower excitation temperature lines are fit by a temperature of 600 ± 100 and a column density of $2 \pm 1 \times 10^{19} \text{ cm}^{-2}$.
5. Based on the wide semiopening angle, high temperature, and low column density for the extended H_2 , we conclude it originates from a disk wind.
6. We derive a molecular wind mass-loss rate of $3 \pm 2 \times 10^{-9} M_\odot \text{ yr}^{-1}$, in line with most literature values but high compared to the previously derived stellar accretion rate of $6.6 \times 10^{-10} M_\odot \text{ yr}^{-1}$. This points to either a photoevaporative origin for the wind or variable accretion onto the central star.

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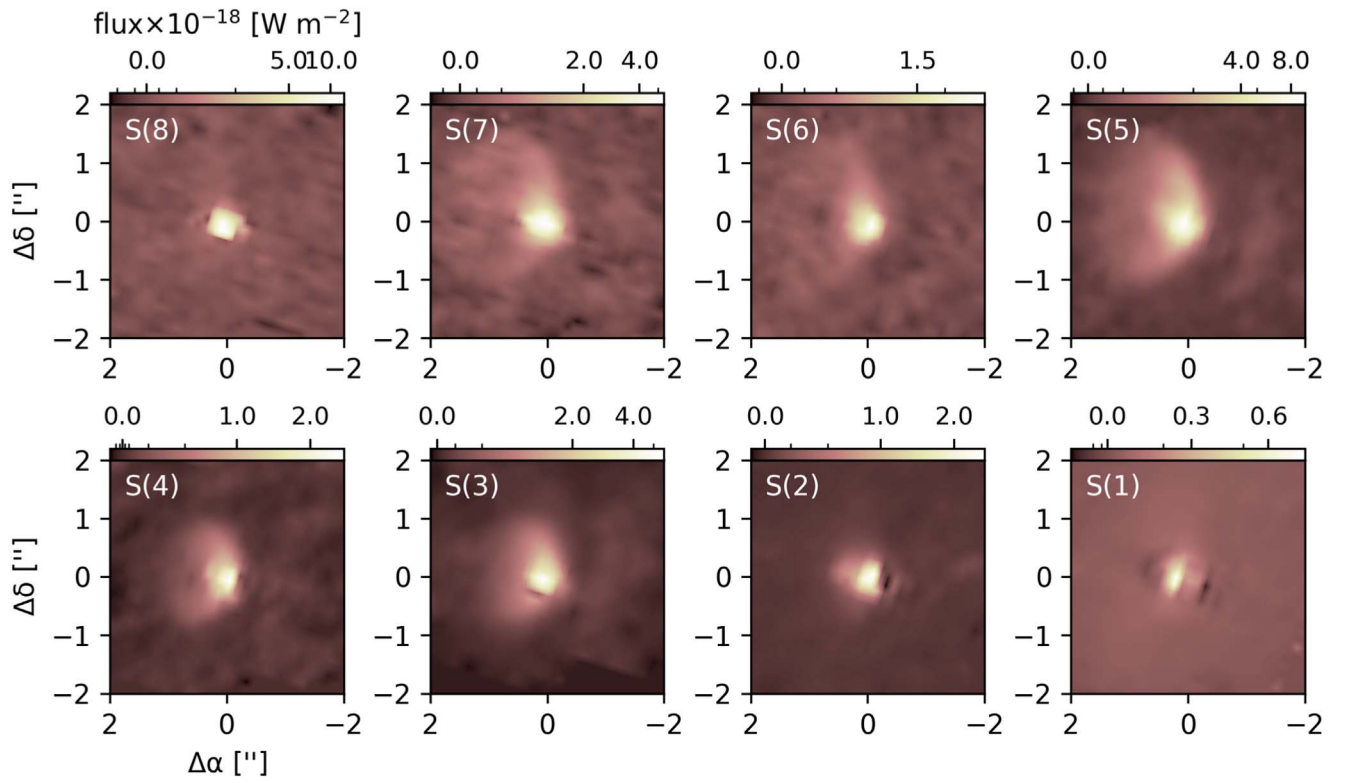
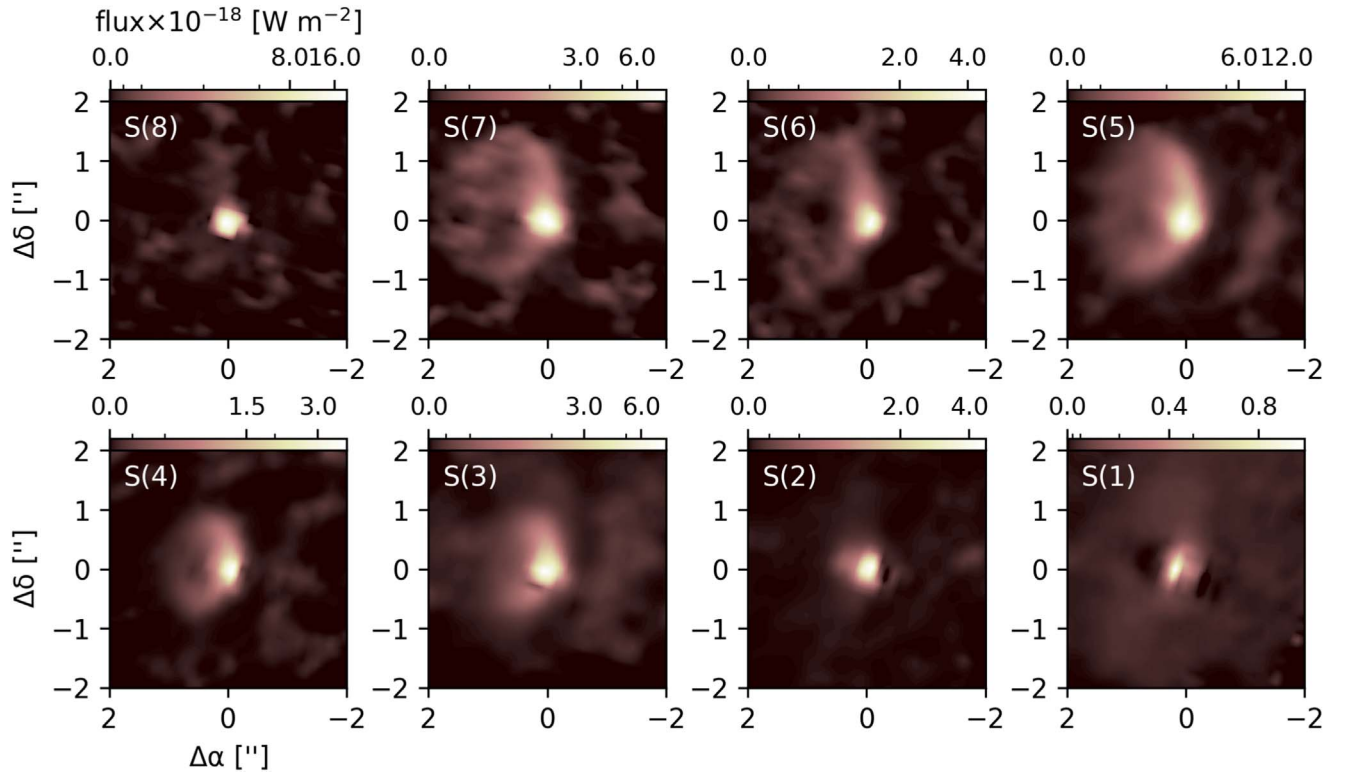
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Facility: JWST.

Software: EMCEE (D. Foreman-Mackey et al. 2013), MATPLOTLIB (J. D. Hunter 2007), MINDS PIPELINE (V. Christiaens et al. 2024), NUMPY (S. van der Walt et al. 2011), SPECTRES (A. C. Carnall 2017), VIP (C. A. Gomez Gonzalez et al. 2017; V. Christiaens et al. 2023).

Appendix A Image Deconvolution

Here we present additional H_2 images: after PSF subtraction but before deconvolution (Figure 5), after 3 rounds of deconvolution (Figure 6), and after 27 rounds (Figure 7).

Figure 5. H_2 lines after PSF subtraction but before deconvolution.Figure 6. H_2 lines after 3 rounds of deconvolution.

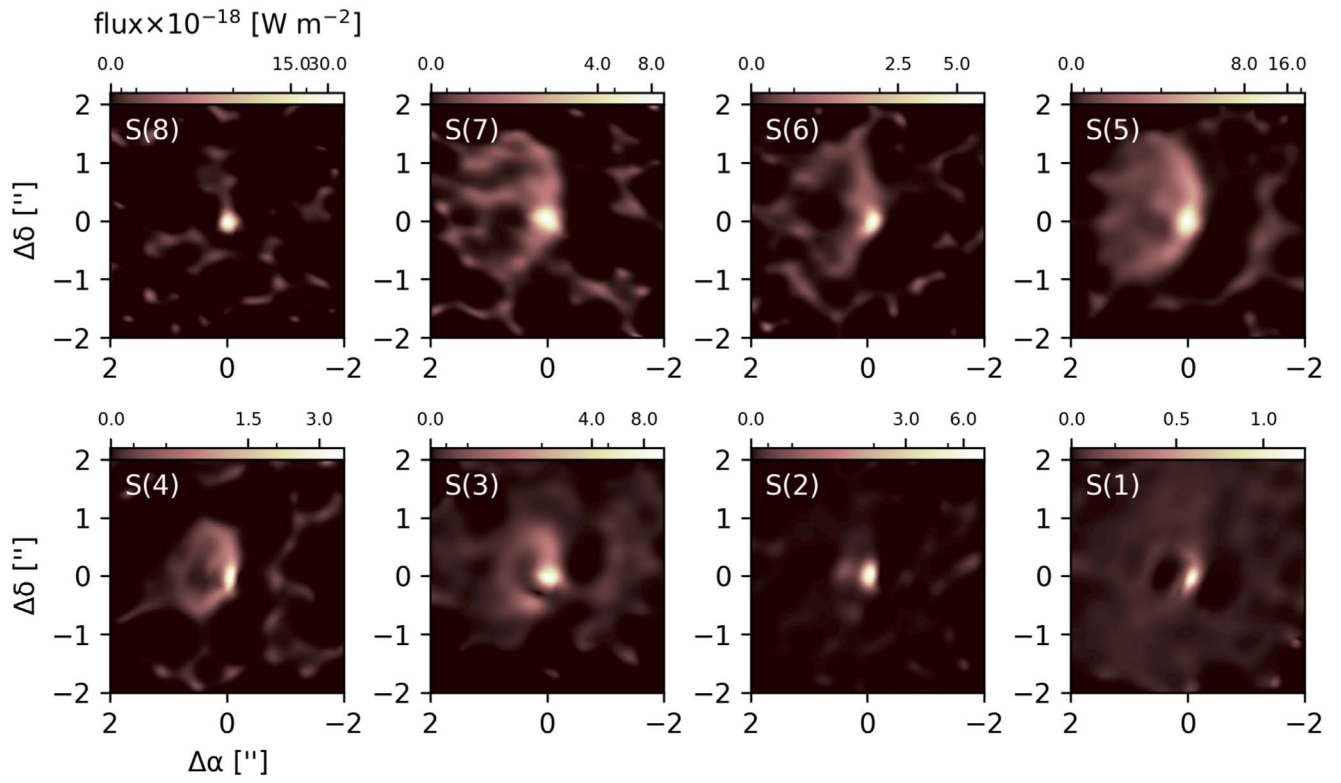


Figure 7. H_2 lines after 27 rounds of deconvolution.

Appendix B Channel Maps

Here we present the channel maps for lines with extended emission in Figures 8–13. These maps have been continuum subtracted using the flux in adjacent channels. No deconvolution has been applied.

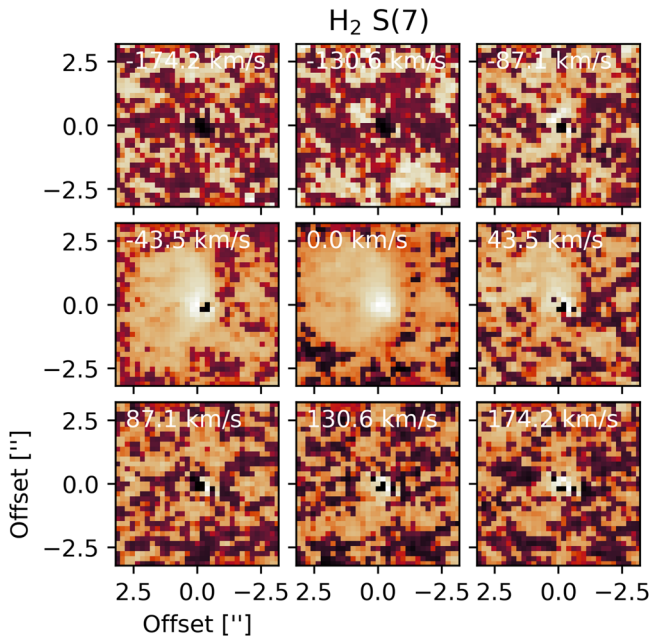


Figure 8. Channel map of the continuum-subtracted data cube for channels surrounding the H_2 S(7) line. Velocity offsets from the line center are shown in the top left of each panel.

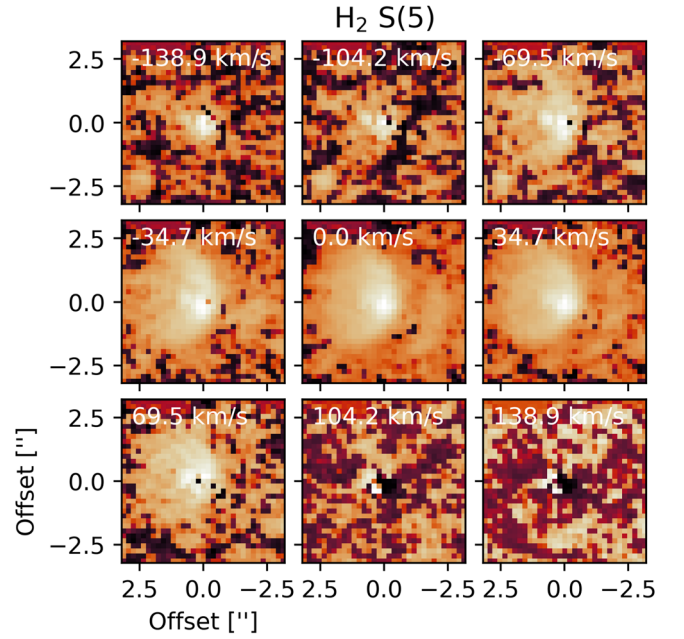


Figure 10. Same as Figure 8 but for H_2 S(5).

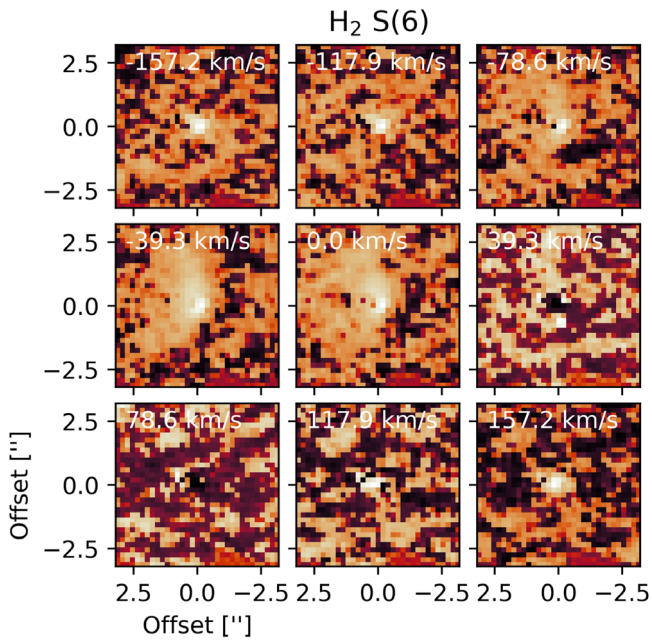


Figure 9. Same as Figure 8 but for H_2 S(6).

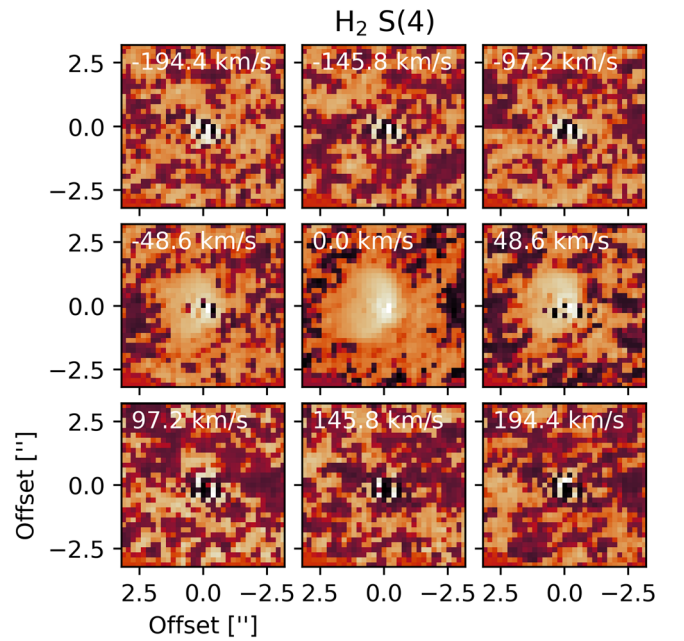


Figure 11. Same as Figure 8 but for H_2 S(4).

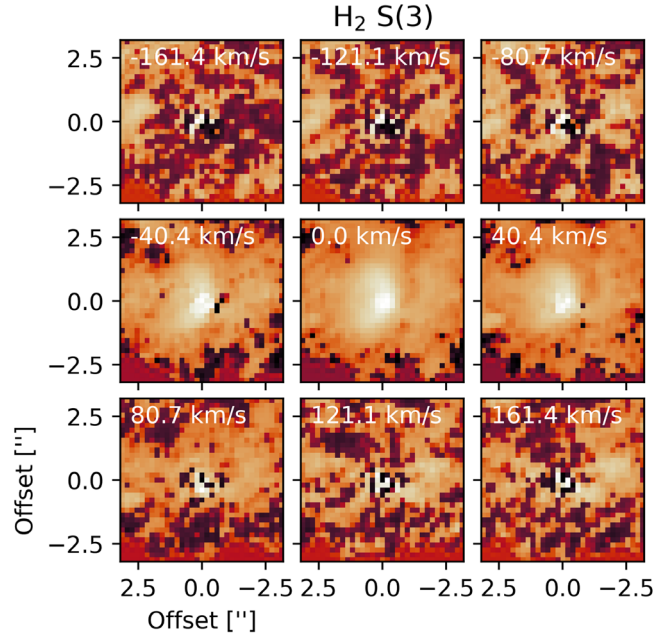


Figure 12. Same as Figure 8 but for H_2 $S(3)$.

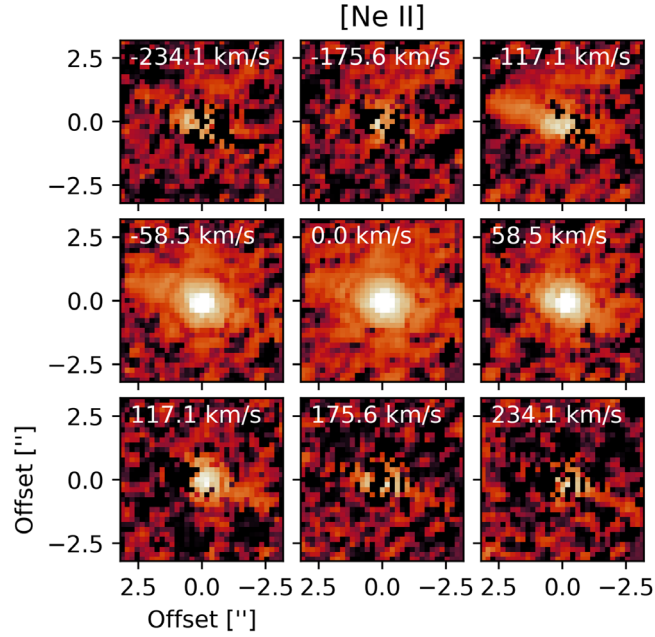


Figure 13. Same as Figure 8 but for $[\text{Ne II}]$. The diffraction pattern seen near 0 km s^{-1} is due to unresolved $[\text{Ne II}]$ emission, which does not follow the subtracted continuum.

Appendix C Fits to Atomic Lines

Here we present the results of our atomic line fits. The local continuum is determined by selecting line-free pixels and interpolating between them using a spline fit. We then fit a

Gaussian line profile to the atomic line using the Markov Chain Monte Carlo package `emcee` (D. Foreman-Mackey et al. 2013). The best-fit models are shown in Figure 14, and the values are given in Table 4.

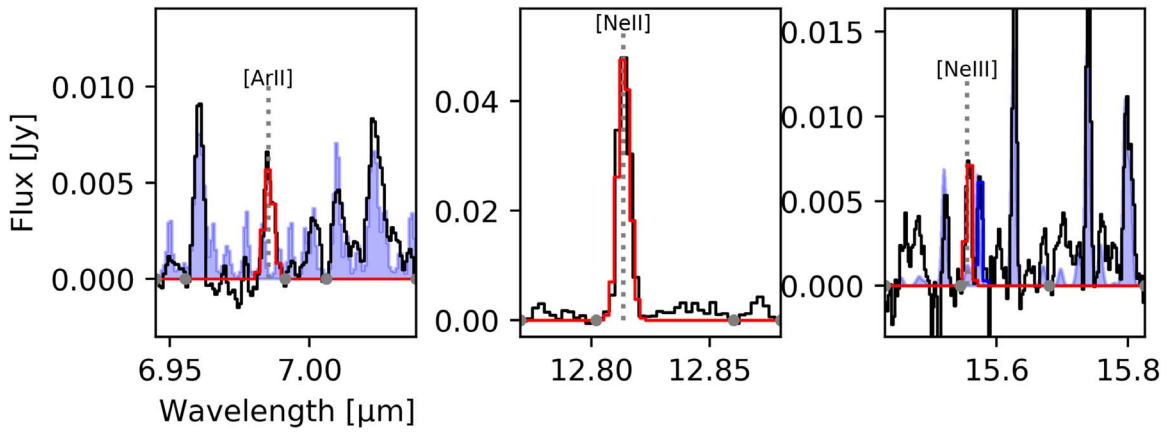


Figure 14. Best-fit solution for the Gaussian fit to the atomic lines. Black is the MRS spectrum after continuum subtraction. Gray dots show the points used for continuum subtraction. The red line is the best fit. In the right panel, the dark-blue line is the best fit to the nearby water line, while the light blue is the best-fit water slab model of K. R. Schwarz et al. (2024).

Table 4
Integrated Flux for Gaussian Fits to Atomic Lines

Transition	Line Center (μm)	σ 10^{-3} (μm)	E_u (K)	n_{crit} (cm^{-2})	Integrated Flux 10^{-15} ($\text{erg s}^{-1} \text{cm}^{-2}$)	rms 10^{-28} ($\text{erg s}^{-1} \text{cm}^{-2} \text{Hz}^{-1}$)
Ar II	6.985	1.7	2059.8	...	4.2e5	6.7
Ne II	12.814	2.6	1122.0	6.3e5	5.87	8.5
Ne III	15.555	4.0	924.9	2.7e5	0.94	9.0

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